Integer Linear Forms for a Factorial Reciprocal Series

An integral basis and a finite gcd formula

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Abstract

For $S = \sum_{n\geq 2}(n! - 1)^{-1}$, we classify the multipliers in cancellation forms $MS + k$ constructed from finite integer vectors supported on $n \geq 2$. An integral basis solves cancellation through depth $D \geq 2$; excluding index one leaves one Diophantine equation. Its coefficient gcd is determined below $2D^2$, giving the least positive multiplier and an attaining vector. At $D = 4$, the minimum is 1380; an attaining form is nonintegral. Irrationality still requires, for each $q > 0$, a nonintegral form with $q \mid M$.

1 Introduction

Erdős asked whether replacing $n!$ by $n! - 1$ preserves the irrationality of the reciprocal sum [1, p. 102]. The question appears as Problem 68 in Bloom's catalogue [2].

Problem 1.1 (Erdős Problem #68)**.** Is

$$S = \sum_{n\geq 2} \frac{1}{n! - 1}$$

irrational?

The elementary proof that e is irrational clears the partial sum by a factorial and bounds the remaining positive tail. Here the first step fails: for $H_m = \sum_{n=2}^{m}(n! - 1)^{-1}$, already $3!H_3 = 36/5$. We instead construct forms $MS + k$, with $M, k \in \mathbb{Z}$, by cancelling initial weighted sums. If $S = a/q$ and $q \mid M$, then $MS+k$ is integral; proving it nonintegral excludes that denominator.

At depth four, the weight congruences force $115 \mid M$. Restricting the coefficient support to $n \geq 2$ strengthens this to $1380 \mid M$, and every multiple of 1380 occurs. We determine the exact factor at every depth $D \geq 2$ and construct a vector attaining the least positive multiplier.

First allow a coefficient at index one. Theorem 2.1 constructs an integral basis by correcting adjacent factorial differences at proper divisors to isolate each weighted sum.

Authorship and AI use. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

At fixed M, cancellation determines the lower basis coefficients; removing index one leaves a single equation in the higher coefficients. Theorem 3.1 reduces its infinite gcd to a calculation below $2D^2$. Bézout coefficients then construct an attaining vector. No assumption on the rationality of S enters this classification.

Hančl and Tijdeman's tail-integrality lemma treats factorial series with integer coefficients [4, Lemma 2.1 and the following remark, p. 385]. Their factorial-scaled partial sums are integers. Here the scaled partial sum need not be integral. Instead, the weight congruences give $\mathcal{R} - MS \in \mathbb{Z}$ (1); the assumption $S = a/q$ and $q \mid M$ then makes $\mathcal{R}$ integral. Koepf and Schmersau's factorial-digit criterion [3, Example 3.2, p. 121] is used in the companion's digit argument to give an equivalent eventual-pattern condition for $S - e + 2$. Failure of that pattern at arbitrarily large indices also remains unproved.

Section 4 tests an attaining form; Section 5 gives the general nonintegrality test.

2 An integral basis

We need integer weights congruent to $n!$ modulo $d! - 1$, with $nW_{d,n-1} = W_{d,n}$ when $n \geq 2$ and $d \nmid n$. Removing powers of $d!$ preserves the residue; the floor exponent gives the required jumps. Put

$$W_{d,n} = \frac{n!}{(d!)^{\lfloor n/d \rfloor}} \qquad (d \geq 2,\ n \geq 1).$$

Writing $n = kd + r$, $0 \leq r < d$, shows that $W_{d,n}$ is $r!$ times the multinomial coefficient $n!/((d!)^k r!)$. Thus it is an integer, and $n! = (d!)^k W_{d,n}$ gives $W_{d,n} \equiv n! \pmod{d! - 1}$. For example, $W_{3,4} = 4$ and $(4! - W_{3,4})/(3! - 1) = 4$.

For a finitely supported integer vector $\lambda = (\lambda_n)_{n \geq 1}$, define the integer linear forms

$$M(\lambda) = \sum_n \lambda_n n!, \qquad V_d(\lambda) = \sum_n \lambda_n W_{d,n},$$

and let

$$\mathcal{R}(\lambda) = \sum_{d \geq 2} \frac{V_d(\lambda)}{d! - 1}.$$

Here n indexes coefficients and d indexes remainder summands. If the support lies in $n \leq N$, $N \geq 2$, then $V_d = M$ for $d > N$. Thus the remainder converges absolutely, and subtracting MS leaves only finitely many terms. The weight congruence makes each an integer:

$$\mathcal{R}(\lambda) - M(\lambda)S = \sum_{d=2}^{N} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} \in \mathbb{Z}. \tag{1}$$

We solve $V_2 = \cdots = V_D = 0$, allowing either sign of M. Write e_n for the unit vector at index n. Index one is auxiliary: it makes the basis triangular but introduces no term $1/(1! - 1)$, since the remainder still starts at $d = 2$. We impose $\lambda_1 = 0$ in Section 3.

The identity $n(n-1)! - n! = 0$ suggests starting with $T_n = ne_{n-1} - e_n$. At $n = 2$, the vector $U_2 = 2e_1 - e_2$ has $V_2 = 1$ and every other weighted sum zero. At $n = 4$, the difference $T_4 = 4e_3 - e_4$ has two nonzero weighted sums. One correction suffices:

Vector	M	V_2	V_3	V_4
T_4	0	6	0	23
$6U_2$	0	6	0	0
$T_4 - 6U_2$	0	0	0	23

For $d > 4$ these weighted sums equal $M = 0$. Thus $U_4 = T_4 - 6U_2$ isolates $V_4 = 4! - 1$. In general, the unwanted sums occur at proper divisors of n, whose correcting vectors are already available in an inductive construction.

Theorem 2.1 (an integer basis with prescribed weighted sums)**.** *Set*

$$T_n = ne_{n-1} - e_n, \qquad U_n = T_n - \sum_{\substack{d \mid n \\ 2 \le d < n}} W_{d,n} U_d \quad (n \ge 2).$$

Then

$$M(U_n) = 0, \qquad V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}.$$

The vectors $e_1, U_2, U_3, \ldots$ form an integral basis. Every finite vector has the unique finite expansion

$$\lambda = M(\lambda)e_1 + \sum_{d \ge 2} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} U_d. \tag{2}$$

Proof. We have $M(T_n) = 0$. If $d \nmid n$, the floor exponents at $n-1$ and n agree, so $nW_{d,n-1} = W_{d,n}$. If $d \mid n$, the exponent increases by one, so $nW_{d,n-1} = d!W_{d,n}$. Hence

$$V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d \mid n}.$$

Suppose the asserted identities hold below n. For each proper divisor d, subtracting $W_{d,n}U_d$ cancels the sum at d and changes none of the others. The sum at n remains $n! - 1$, since $W_{n,n} = 1$. This proves the identities by induction.

For completeness, restrict to indices $1, \ldots, N$. The columns $e_1, T_2, \ldots, T_N$ form a triangular integer matrix with diagonal $1, -1, \ldots, -1$. Replacing each T_n by U_n subtracts an integer combination of earlier columns, so the second change of basis is triangular with diagonal one. Both matrices have integer inverses: back-substitution divides only by 1 or -1. Hence $e_1, U_2, \ldots, U_N$ is a basis over $\mathbb{Z}$.

In the resulting expansion, applying M gives the coefficient of e_1. If the coefficient of U_d is c_d, applying V_d gives $V_d = M + (d! - 1)c_d$, which proves (2). For $d > N$ that coefficient is zero because $V_d = M$. □

The U_d-coordinates in (2) are exactly the integer summands in (1). Their sum is therefore k in $\mathcal{R} = MS + k$.

For a prime $p \ge 3$ the divisor sum is empty, and $U_p = pe_{p-1} - e_p$. At a composite index the correction terms remove the extra weighted sums. For example,

$$U_9 = 9e_8 - e_9 - 5040e_2 + 1680e_3. \tag{3}$$

Adding U_n changes only V_n among the weighted sums, preserves M, and adds one to $\mathcal{R}$. Integer multiples therefore preserve the fractional part.

3 Possible values of M

Fix $D \geq 2$ and solve $V_2 = \cdots = V_D = 0$, allowing index one temporarily. The coordinates in (2) give

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1}. \tag{4}$$

It follows that M must be divisible by

$$L_D = \operatorname{lcm}_{2 \leq d \leq D}(d! - 1).$$

This necessary condition is sufficient while index one is allowed: $L_D e_1$ has every weighted sum equal to L_D, and subtracting $L_D/(d! - 1)$ copies of U_d sets the dth sum to zero without changing the others. Thus put

$$K_D = L_D e_1 - \sum_{d=2}^{D} \frac{L_D}{d! - 1} U_d.$$

Uniqueness of the coordinates now gives the complete solution

$$V_2 = \cdots = V_D = 0 \quad \Longleftrightarrow \quad \lambda = tK_D + \sum_{n>D} z_n U_n, \qquad M = tL_D, \tag{5}$$

with $t \in \mathbb{Z}$ and finitely many nonzero integers z_n. The coefficient of e_1 in (2) is M; the original coordinate λ_1 also receives contributions from the U_n. Write $a_D = (K_D)_1$ and $u_n = (U_n)_1$. The support restriction is exactly

$$ta_D + \sum_{n>D} z_n u_n = 0. \tag{6}$$

The higher corrections leave cancellation untouched and change the first coordinate by exactly $g_D\mathbb{Z}$, where $g_D = \gcd\{u_n : n > D\}$. They can therefore remove ta_D precisely when $g_D \mid ta_D$. For example, at depth four we shall find $a_4 = -55$ and $g_4 = 60$. Removing index one then requires $60 \mid 55t$, or $12 \mid t$; hence $M = 115t$ must be a multiple of 1380.

In general, the possible values of M form the ideal

$$\mu_D\mathbb{Z}, \qquad \mu_D = L_D \frac{g_D}{\gcd(g_D, a_D)}. \tag{7}$$

Indeed, division by $\gcd(g_D, a_D)$ leaves two coprime integers, so $g_D \mid ta_D$ holds exactly when $g_D/\gcd(g_D, a_D) \mid t$. Bézout's identity supplies the required finite combination; the next theorem bounds the indices needed. Every vector attaining μ_D has coefficient gcd one, since division by a common factor would give a smaller positive multiplier.

3.1 A finite gcd calculation

The recursion for the first coordinate is

$$u_2 = 2, \qquad u_n = -\sum_{\substack{d \mid n \\ 2 \leq d < n}} W_{d,n} u_d \quad (n > 2).$$

Induction gives $u_n = 0$ at odd indices and $u_{2p} = -(2p)!/2^{p-1} \neq 0$ at twice a prime, including $p = 2$. Choose a prime ℓ with $D/2 < \ell \leq D$, using Bertrand's postulate for $D \geq 3$ and $\ell = 2$ for $D = 2$. A finite gcd containing $u_{2\ell}$ is positive and divides $(2\ell)!$.

The remaining issue is whether later coefficients reduce this gcd. In each recurrence term $W_{d,n}u_d$, induction controls u_d for $d > D$. For $d \leq D$, unordered block counting gives $(n/d)! \mid W_{d,n}$. The cutoff ensures $n/d \geq 2\ell$, so the weight supplies the factor $(2\ell)!$ instead.

Theorem 3.1 (a finite formula for the gcd)**.** *Choose a prime ℓ with $D/2 < \ell \leq D$ and put $H = D(2\ell - 1)$. Then*

$$g_D = \gcd(u_{D+1}, \dots, u_H), \qquad H < 2D^2.$$

Proof. Let $g = \gcd(u_{D+1}, \dots, u_H)$. The index 2ℓ belongs to this interval, and $u_{2\ell} = -(2\ell)!/2^{\ell-1} \neq 0$. Consequently $g > 0$ and $g \mid (2\ell)!$.

We prove $g \mid u_n$ for every $n > D$ by strong induction. This is true up to H by definition. For $n > H$, consider a proper divisor $d \geq 2$ in the recurrence for u_n. If $d > D$, the induction hypothesis gives $g \mid u_d$. If $d \leq D$, put $k = n/d$. The choice $H = D(2\ell - 1)$ ensures $k \geq 2\ell$. Moreover,

$$\frac{W_{d,n}}{k!} = \frac{n!}{(d!)^k k!} \in \mathbb{Z} :$$

this quotient counts partitions of n labelled objects into k unordered blocks of size d. Hence $(2\ell)! \mid k! \mid W_{d,n}$, so $g \mid W_{d,n}$ in the small-divisor case as well. Every product $W_{d,n}u_d$ in the recurrence is therefore divisible by g.

Thus g divides the whole tail. Conversely, every common divisor of the tail divides its finite subfamily $u_{D+1}, \dots, u_H$. The two gcds are equal, and $H \leq D(2D-1) < 2D^2$. □

The coefficient at $2\ell \leq 2D$ supplies positivity; the larger cutoff controls the small-divisor weights. The argument does not justify truncation at $2D$.

To attain the least positive multiplier, compute integers b_n with $\sum_{n=D+1}^{H} b_n u_n = g_D$ by the extended Euclidean algorithm, and put $h = \gcd(g_D, a_D)$. Taking

$$t = g_D/h, \qquad z_n = -(a_D/h)b_n \quad (D < n \leq H)$$

makes $ta_D + \sum z_n u_n = 0$. Substitution in (5) therefore gives $M = \mu_D$ with support in $2, \dots, H$.

3.2 The associated remainder

Fix $M \in \mu_D\mathbb{Z}$ and put $t = M/L_D$. For choices of z_n satisfying (6), the identities $\mathcal{R}(e_1) = S$ and $\mathcal{R}(U_n) = 1$ give

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n, \qquad H_D = \sum_{d=2}^{D} \frac{1}{d! - 1}. \tag{8}$$

Since $MH_D \in \mathbb{Z}$, the offset from MS is $k = -MH_D + \sum_{n>D} z_n$; $\sum z_n$ alone is the offset from $M(S - H_D)$. All choices at fixed M therefore have the same fractional part.

4 The least multiplier at depth four

At depth four the recursion gives

$$L_4 = 115, \qquad K_4 = -55e_1 + 16e_2 + 3e_3 + 5e_4, \qquad a_4 = -55.$$

Take $\ell = 3$ in Theorem 3.1, so that $H = 20$. Since $u_6 = -180$, $u_8 = -4200$ and $23u_6 - u_8 = 60$, we have $g_4 \mid 60$. For the converse,

$$\operatorname{lcm}(1, \dots, n) \mid u_n \tag{9}$$

gives $60 \mid u_n$ for every $n > 4$, hence $60 \mid g_4$. To prove (9), put $\Lambda_n = \operatorname{lcm}(1, \dots, n)$. For a prime r, write v_r for its valuation. If $d \mid n$, Legendre's formula gives

$$v_r(W_{d,n}) = \sum_{j \geq 1} \left(\left\lfloor \frac{n}{r^j} \right\rfloor - \frac{n}{d} \left\lfloor \frac{d}{r^j} \right\rfloor \right) \geq v_r(\Lambda_n) - v_r(\Lambda_d).$$

Each summand is nonnegative, and each power $d < r^j \leq n$ contributes at least one. Hence $\Lambda_n \mid W_{d,n}\Lambda_d$, and induction in the recurrence, starting from $u_2 = 2$, proves the assertion. Thus $g_4 = 60$, and (7) gives

$$\mu_4 = 115 \frac{60}{\gcd(60, 55)} = 1380.$$

To attain $M = 1380$, take $t = 12$ in (6). The choice $z_6 = -27$, $z_8 = 1$ works because

$$12(-55) - 27(-180) - 4200 = 0.$$

The vector $12K_4 - 27U_6 + U_8$ therefore has no index-one coefficient and cancels V_2, V_3, V_4. To remove its coefficients at indices four and seven, add $-26U_5 + 8U_7$. These corrections still cancel through four and do not restore index one, since $u_5 = u_7 = 0$. Thus

$$\lambda = 12K_4 - 26U_5 - 27U_6 + 8U_7 + U_8,$$

which expands to

$$\lambda = 1482e_2 - 784e_3 - 136e_5 + 83e_6 - e_8, \qquad (M, V_2, V_3, V_4) = (1380, 0, 0, 0). \tag{10}$$

Table 1 verifies all four sums directly.

The basis coordinates give the offset $-26-27+8+1 = -44$ in (8). Since $H_4 = 143/115$, the original linear form is

$$\mathcal{R}(\lambda) = 1380(S - H_4) - 44 = 1380S - 1760.$$

For the remainder estimate, the inequality $(n+1)!-1 > (n+1)(n!-1)$ gives the geometric bound

$$0 < S - H_N < \frac{N+2}{(N+1)((N+1)!-1)} < \frac{2}{(N+1)!-1} \qquad (N \geq 2). \tag{11}$$

The majorant starts with $1/((N+1)!-1)$ and has ratio $1/(N+2)$.

n	λ_n	$n!$	$W_{2,n}$	$W_{3,n}$	$W_{4,n}$
2	1482	2	1	2	2
3	-784	6	3	1	6
5	-136	120	30	20	5
6	83	720	90	20	30
8	-1	40320	2520	1120	70
Weighted totals		1380	0	0	0

Table 1: The depth-four checks: multiply each of the last four columns by λ_n and sum to obtain the last row.

Exact rational arithmetic gives

$$-31+\frac{4}{5}<1380\sum_{d=5}^{8}\frac{1}{d!-1}-44<-31+\frac{5}{6},\qquad \frac{2\cdot 1380}{9!-1}<\frac{1}{100}.$$

The omitted contribution to $\mathcal{R}$ is positive and less than $1/100$, so $-31<\mathcal{R}(\lambda)<-30$. If $S=a/q$ with $q\mid 1380$, (1) would make this remainder an integer. Every divisor of 1380 is therefore excluded.

The depth-six calculation gives $\mu_6=24L_6$: the extra factor need not equal 12. The companion also gives a dual congruence, support bounds and coefficient-norm refinements. Least multiplier, least support size and least coefficient norm are distinct problems; (7) settles only the first.

5 Nonintegrality of the remainder

Fix a vector with $V_2=\cdots=V_D=0$ and $M>0$, and choose any cutoff $N\ge D$ at or beyond its support. Separate the signed finite sum from the positive tail:

$$A_N=\sum_{d=D+1}^{N}\frac{V_d(\lambda)}{d!-1},\qquad \mathcal{R}(\lambda)=A_N+M(S-H_N).$$

The remainder exceeds A_N. By (11), it lies below the least integer strictly above A_N whenever

$$\frac{2M}{(N+1)!-1}<\lfloor A_N\rfloor+1-A_N. \tag{12}$$

Then $\lfloor A_N\rfloor<\mathcal{R}(\lambda)<\lfloor A_N\rfloor+1$, excluding every denominator dividing M. The strict successor matters: at an integral A_N the gap is one, whereas just below an integer it can be arbitrarily small. Failure of the comparison is inconclusive.

At depth two, fix $\lambda=-6e_2+e_4$, with $M=12$, and vary only the cutoff. At $N=4$, its finite part $A_4=-239/115$ lies just below -2, and its gap $9/115$ is smaller than the bound $24/119$. At $N=5$, the finite part $A_5=-27061/13685$ has crossed -2: the next integer is now -1, and the gap $13376/13685$ exceeds $24/719$. For any fixed vector with $M>0$, A_N increases to $\mathcal{R}$. If $\mathcal{R}$ is nonintegral, the gap tends to a positive number and

the test eventually succeeds. If $\mathcal{R}$ is integral, the eventual gap equals the omitted tail, so the strict upper-bound test fails. Changing the cutoff detects nonintegrality; it neither creates it nor changes M. Coefficient changes at fixed M preserve nonintegrality by (8).

To exclude every denominator, M must vary. The companion's progression construction adds size control: polynomial root conditions give primitive vectors (coefficient gcd one) with $0 < M < N!$, where N is their largest supported index. Every fixed q divides their multipliers for all sufficiently large N. The tail at that endpoint is below $2/N$, but the next-integer gap may shrink too. For each $q > 0$, one still needs a vector with $q \mid M$ and a cutoff satisfying (12); the cutoff may exceed its support endpoint. No minimal-support assertion is made.

The companion treats the equivalent carry condition and the complementary-denominator tail inequality. It also records two larger finite computations: a non-unit carry at $m = 300000$ excludes every $q \mid 299999!$, and a continued-fraction enclosure excludes every $q < 2^{39990}$. Both computations were performed outside Lean; their algorithms, exact outputs and scope are given there. The factorial-divisibility exclusion does not exclude all denominators with only small prime factors, whose exponents may be larger. Neither finite restriction establishes the required condition at arbitrarily large indices.

6 Verification

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

The integral-basis theorem and finite gcd formula have recorded exact Lean bindings. The companion source concordance identifies their declarations and the theorem on the possible values of M. Its large finite computations are separate from these kernel proofs.

Acknowledgements

An AI research pass supplied by Will Cook derived the depth-four dual congruence and shorter vectors from the weighted linear forms and the classification of their possible coefficients M developed earlier here. OpenAI Codex checked and integrated them; a separate AI pass reviewed the proof.

References

[1] Paul Erdős. On the irrationality of certain series: problems and results. In Alan Baker (ed.), *New Advances in Transcendence Theory*, Cambridge University Press (1988), pp. 102–109.

[2] Thomas F. Bloom. Erdős Problem #68. Online resource (2026). Historical access: 28 July 2026; present-page status not reverified.

[3] Wolfram Koepf and Dieter Schmersau. Irrationality of certain infinite series II. *Analysis* **31** (2011), 117–124.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 2.1	Lean · Comparator	Theorem 3.1	Lean · Comparator